

B.C.S.E. FINAL YEAR EXAMINATION, 2017

(First Semester)

Pattern Recognition & Image Processing

Time : 3 hours

Full marks : 100

Answer question no.1 and any four from the rest

All questions carry equal marks

1. Answer true or false stating reasons

2x10 = 20

- (a) Partitional clustering methods help in exploring data at different levels of granularity.
- (b) Kohonen self-organizing network can be used for data coding.
- (c) The success of a pattern classification scheme using decision function depends only on finding the appropriate form of $d(x)$.
- (d) It is not analytically possible to compare the clustering results of two different clustering algorithms.
- (e) The measurement unit used cannot affect the clustering result of a given data set.
- (f) Each node in a typical ANN can have any number of outgoing connections, where the signals in all of these may not be the same.
- (g) In a typical PR problem, the dimensionality of measurement space is either greater than or less than that of feature space.
- (h) Image enhancement techniques are objective, in the sense that they tend to be based on mathematical or probabilistic models of image degradation.
- (i) The *Tanimoto measure* is used to find a measure of dissimilarity between patterns where the features are having binary values.
- (j) Segmentation deals with the tools for extracting image components that are useful in the representation and description of shape.

2. (a) Explain briefly the necessary steps involved in a typical Pattern Classifier.

6

- (b) Design a classifier that can classify a set of given text documents into a number of classes depending on their contents where the number of such classes is known *a priori*. Comment on the merits and demerits of such a classifier. 12 + 2

3. What is partitional clustering? State and explain one such method which is based on the principle of minimization of intra-class distances. Discuss the merits and demerits of the method. 2 + 12 + 6

4. (a) Describe how the features of a biological neurone are being imitated in an artificial neurone. 6

- (b) Describe briefly the Kohonen's Self Organizing Network. How such a network can be employed to solve some typical problems of PR. 6 + 8

5. Suppose that a movie production house wants to assess the possible profit of one of their movies which would be launched in near future. Assume that the said production house is having a database containing the following information regarding each of several hundred movies of the past.

- (i) Rating of basic film genres such as action, adventure, animation, biography, comedy etc.
- (ii) Rating of the Production House
- (iii) Rating of popularity of leading actor/actress at the time of casting
- (iv) Total profit/loss

Note that a movie has a value of rating for each genre where the said values would depend on the type of the movie. For instance, an action movie would have a very a high value of rating for 'action' genre, but would have a non-zero positive value for others genres also such as say 'comedy' genre depending upon the extent of comedy present in the movie.

- (a) Formulate the problem as a data mining problem
- (b) Describe the necessary steps involved in solving the problem
- (c) Discuss the merits and demerits of the method you have suggested

4 + 12 + 4

6. What is image segmentation? Briefly describe the three basic approaches in segmenting an image. Suppose you do not have any prior information about the number of objects or segments in a given image. Describe a segmentation method that can extract the appropriate segments of the image.

2 + 8 + 10

7. Write short notes on any two of the following.

2 X 10

- (a) Importance of non-metric similarity measure in clustering
- (b) A typical Image Processing System
- (c) Isodata algorithm